ESE 2113 : Statistics for Environment

MODULE 01:
Environmental Data and Frequency Distributions

Detailed lecture and self-study edition

Second Year, First Semester • Excel + R

SYLLABUS:

ESE 2113. Statistics for Environment **Credits: 3; Contact Hours: 3 Hours/Week**

Frequency distribution: Measures of central tendency, dispersion; moments, skewness and Kurtosis. **Correlation and Regression:** Bivarate data, correlation and regression coefficients, regression. Probability: Sample space event, probability of events, theorem, of total and compound probability, conditional probability, bayes theorem. **Random variable:** Probability function, distribution function, joint, marginal and conditional probability function, expectation and moment generating function. **Parent Distributions:** Binomial, poisson, negative bionomial, normal and exponential distribution. **Sampling Distributions:** c^2, t and F distributions. **Test of Hypothesis:** Test for population, mean, variance, correlation coefficient and regression coefficients.

Recommended References:

- Spiegel M R, Theory and Problems of Statistics
- Hoel P G, Elementary Statistics
- Hogg R V & Craig A T, Introduction to Mathematical Statistics, 4th Ed, Macmillan, London
- Mood, Graybil & Boes, Introduction to the Theory of Statistics, 3rd Ed, McGraw Hill, NY
- Mostafa M G, Methods of Statistics
- Milnyk M, Principles of Applied Statistics
- Barlow R G, Statistics
- Loveday, Practical Statistics and Probability
- Mosteller, Rourk & Thomas, Probability with Statistical Applications, 2nd Ed, Addison-Wesley, USA
- Goon A M & Gupta M N, Fundamentals of Statistics, Vol-I

What is Statistics?

Statistics is the science of collecting, organizing, presenting, analyzing and interpreting data to understand variation and draw conclusions under uncertainty.

Origin of the term

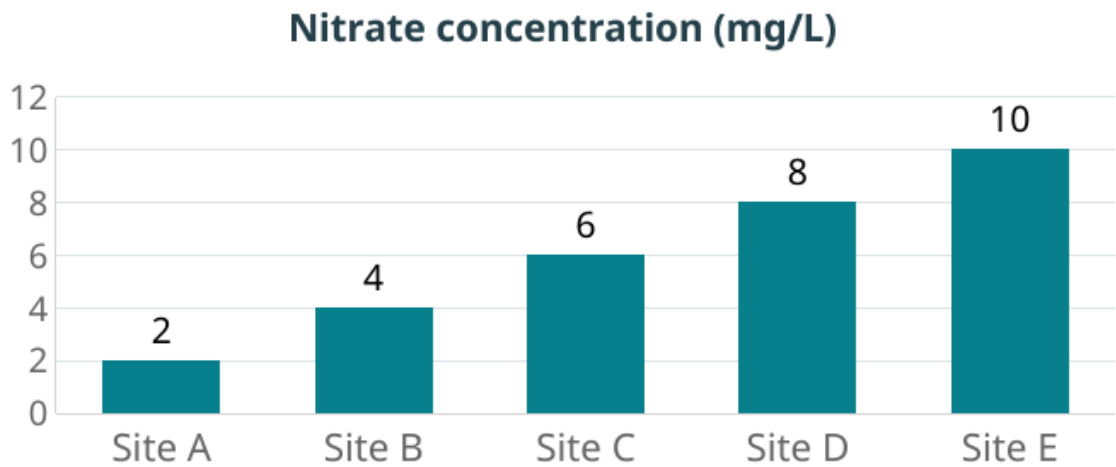
The word comes from German Statistik, originally concerned with information about the state. Its roots connect to Latin status and Italian statista (statesman).

Gottfried Achenwall popularized the term in the 18th century. His 1749 textbook described European states.

Two main branches

Descriptive: summarizes observed data.
Inferential: uses sample data to draw conclusions about a population, with uncertainty.

Figure: river nitrate measurements



Sample mean = (2 + 4 + 6 + 8 + 10) / 5 = 6 mg/L

The mean is a statistic: a numerical summary calculated from a sample.

Statistics in an environmental investigation

Descriptive statistics summarize the observations actually collected. A histogram of 20 measurements describes those 20 results.

Inferential statistics use a sampling design and probability model to learn about a larger population. A site mean does not automatically represent the entire city.

A defensible analysis connects the question, target population, sampling design, measurement method and uncertainty.

Example: “What fraction of daily PM2.5 values at one station exceed a specified study threshold?” differs from “What is the citywide annual exposure?”

Population, sample and observational unit

Term	Definition	Environmental example
Population	All units targeted by the question	All station-days during a defined year
Sample	Observed subset of those units	20 selected station-days
Observational unit	Unit contributing one independent result	One station on one selected day
Parameter	Numerical population characteristic	Annual mean concentration μ
Statistic	Numerical sample characteristic	Mean of the sampled days $\bar{x}$

Define geography, time window, sampling unit and units before calculating.

Independent samples and technical replicates

Five bottles from five independently chosen river locations provide spatial information. Five aliquots from one bottle primarily characterize laboratory precision.

Treating aliquots as five independent river sites creates pseudoreplication and understates uncertainty.

Repeated measurements at one station may be temporally correlated. Record site ID and sampling time so dependence can be assessed.

A larger number of readings does not repair biased site selection.

Variable types and measurement scales

Scale	Meaning	Example / valid operations
Nominal	Categories without order	Land use: industrial, residential
Ordinal	Ordered categories, unequal gaps possible	Pollution class: low, medium, high
Interval	Equal intervals, arbitrary zero	Temperature in °C: differences meaningful
Ratio	Equal intervals and meaningful zero	Mass in mg: ratios meaningful
Discrete / continuous	Countable / continuum of values	Particle count / concentration

Temperature of 20°C is not twice as hot as 10°C. A mass of 20 mg is twice 10 mg.

Raw data and a data dictionary

Each row should describe one observation. Each column should have a unique variable name, unit and definition.

Record site, date, depth, method, replicate type and detection limit alongside the measurement.

Keep missing results as missing with a reason code. Zero, missing and “below detection” describe different situations.

The 20 PM2.5 values in this module are invented for teaching. No regulatory claim follows from them.

Frequency distribution: the components

A frequency f_i counts observations within class i. The sum of all class frequencies must equal n.

Relative frequency $r_i = f_i/n$ gives a proportion. Percentage frequency is $100r_i$.

Cumulative frequency F_i adds all frequencies through class i. The last cumulative frequency equals n.

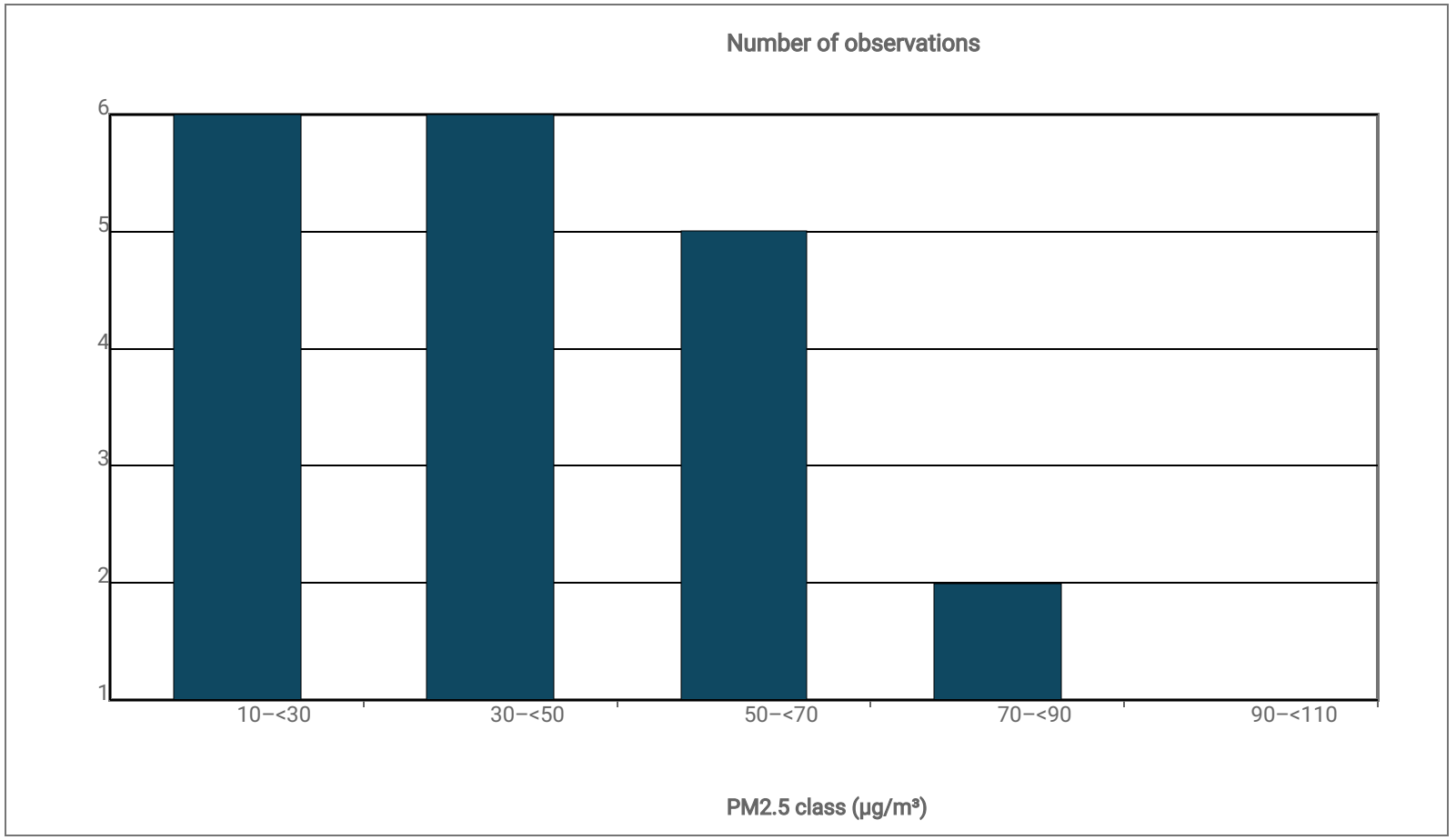
Use nonoverlapping, exhaustive classes. The rule $10 \le x < 30$ includes 10 but excludes 30.

Worked PM2.5 frequency table

Class (µg/m³)	Midpoint	f	Relative f	Cumulative f
10 to <30	20.0	6	0.30	6
30 to <50	40.0	6	0.30	12
50 to <70	60.0	5	0.25	17
70 to <90	80.0	2	0.10	19
90 to <110	100.0	1	0.05	20

Data: 12,18,21,24,27,29,32,35,38,42,44,47,51,54,58,63,68,72,85,96. n=20.

Histogram: frequency within equal-width classes



How to read this graph

Each bar represents a 20 µg/m³ interval.

12/20 observations are below 50 µg/m³.

The upper classes form a relatively sparse right tail.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Class limits, boundaries and midpoints

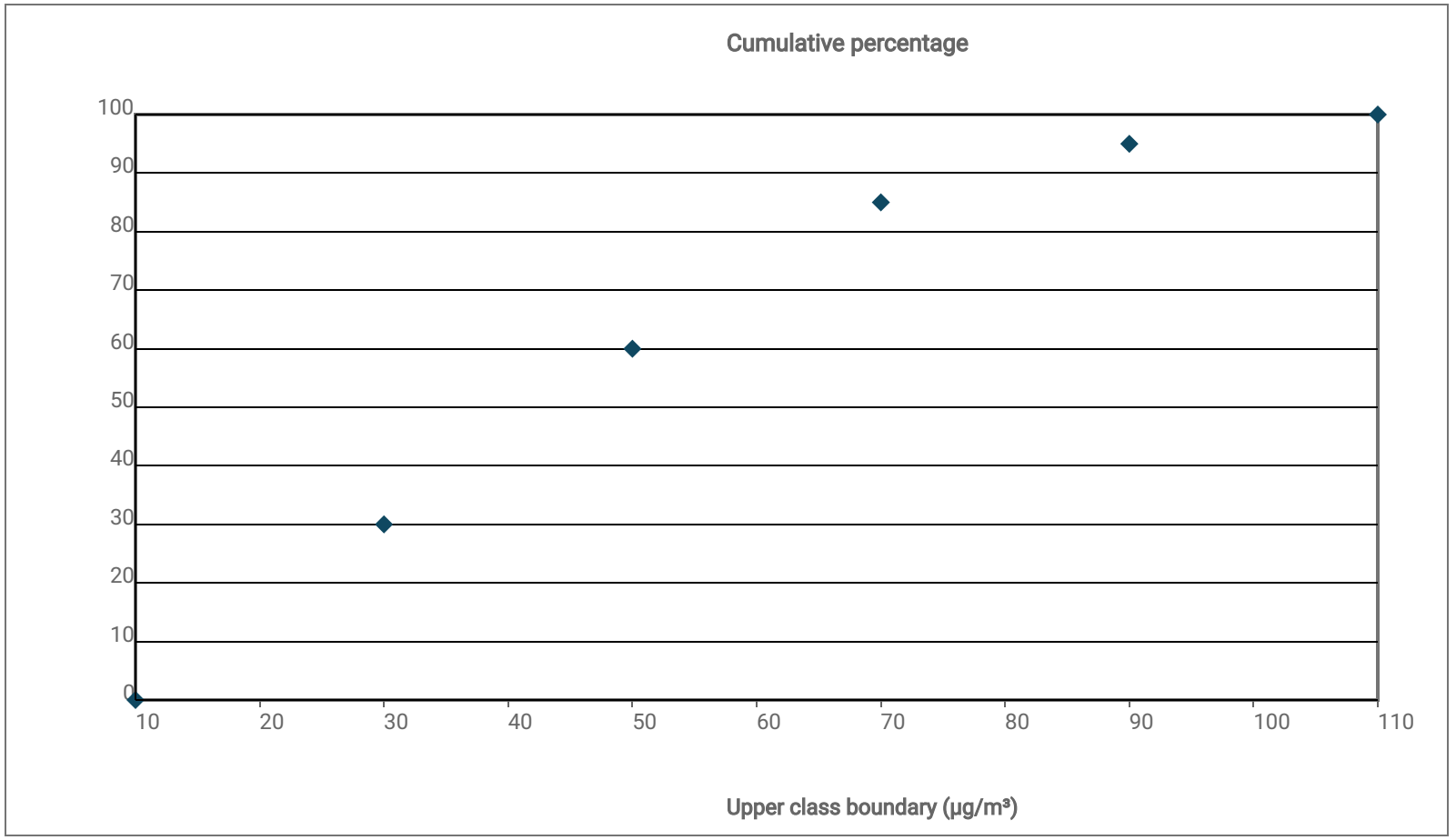
Class limits describe the included values. For continuous data use explicit boundaries such as [10,30).

If values are recorded to the nearest integer, limits 10–29 correspond to measurement boundaries 9.5–29.5. Do not mix this convention with [10,30).

Midpoint = (lower boundary + upper boundary)/2. It represents all observations in a grouped calculation.

Grouping loses exact within-class positions. A grouped mean or percentile is usually an approximation.

Ogive: cumulative percentage below a boundary



How to read this graph

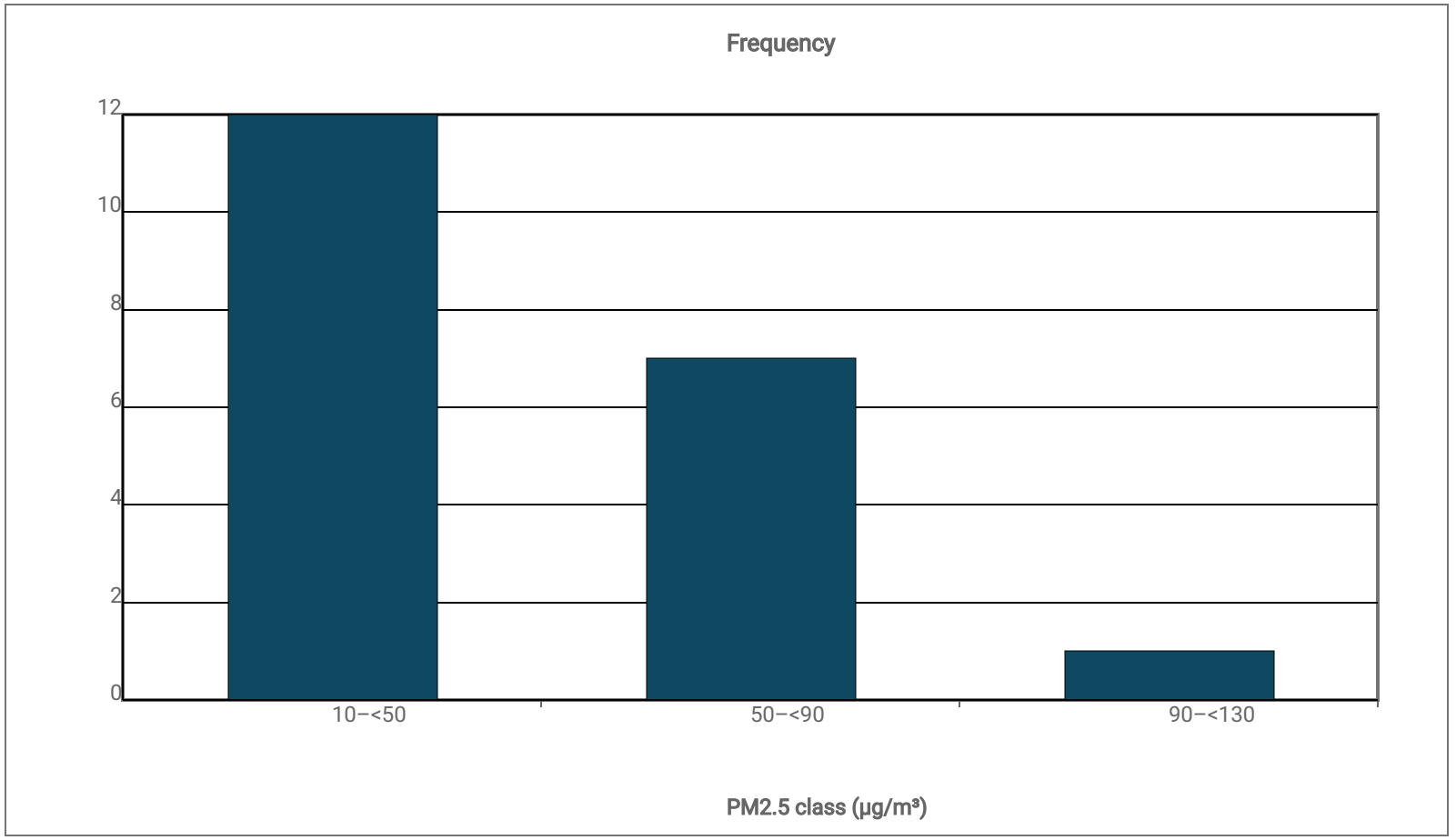
The curve rises and never falls.

At boundary 70, 85% of the sample lies below it.

Interpolation within a class assumes an approximately uniform within-class spread.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Changing class width changes the visual pattern



Illustrative data or theoretical distribution. Exact plotted values remain editable.

How to read this graph

These are the same 20 measurements in wider classes.

Wider bins smooth away local detail.

Compare alternative widths before describing a peak or gap.

Choosing the number of classes

A practical first view often uses roughly √n bins. Sturges' rule gives $k \approx 1 + \log_2 n$. Neither is a universal law.

The Freedman–Diaconis width is h = 2 IQR n^(−1/3). It responds to spread and sample size.

Choose readable boundaries, then check whether a different bin width changes your interpretation.

With a small dataset, show the raw values or a dot plot in addition to a grouped table.

Unequal class widths require frequency density

For unequal widths, bar height should be f_i / class width when area represents frequency.

Suppose [0,10) contains 5 values and [10,30) contains 10. Heights are 0.5 and 0.5, even though frequencies differ.

Using frequencies directly would make the wider interval look artificially more concentrated.

A probability-density histogram uses $f_i/(n \times \text{width})$, so total bar area equals 1.

Frequency and cumulative probability | Try it

Use the worked table. What proportion is in [50,70)?

What proportion is at least 70?

Why can the table not give the exact number above 60?

Frequency and cumulative probability | Solution

[50,70): 5/20 = 0.25, or 25%.

At least 70: (2+1)/20 = 0.15, or 15%.

The class [50,70) crosses 60. Raw data are needed to split it exactly.

A complete frequency analysis from raw observations

Result	Calculation	Meaning
Sample size	n=20	Twenty illustrative station-days
Below 50	12/20=60%	Sample cumulative proportion
At least 70	3/20=15%	Sample upper-tail proportion
Exact mean	$\Sigma x/n=916/20=45.8$	Uses individual measurements
Grouped mean	920/20=46	Midpoint approximation

Grouped numerator=6×20+6×40+5×60+2×80+1×100=920. Exact values remain preferable when available.

Sampling bias in a frequency table

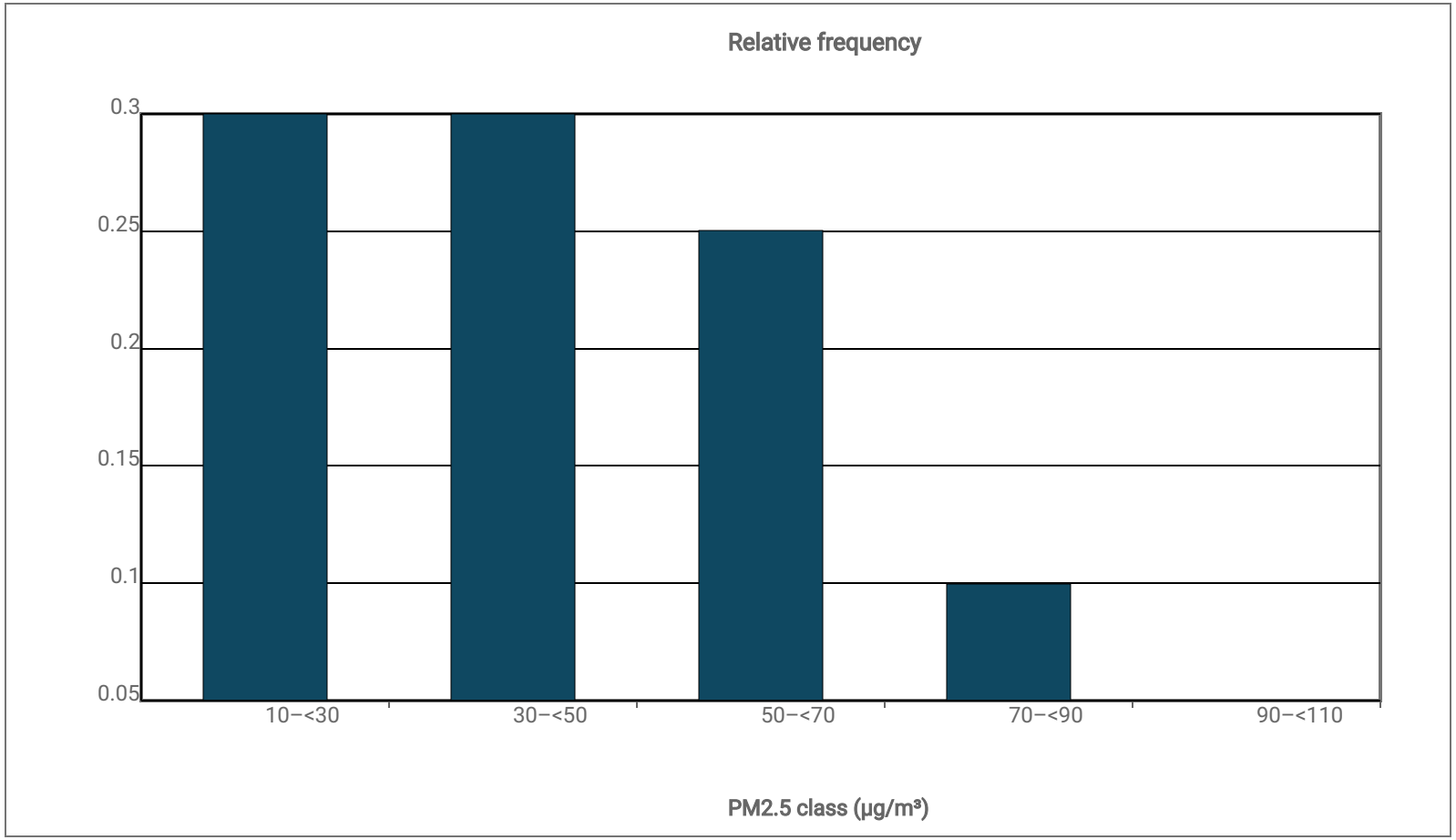
A table may be arithmetically correct and still describe the wrong population. Sampling only weekdays misses weekend behavior.

A roadside station overrepresents local traffic influence if used to estimate citywide background concentrations.

Missing measurements during instrument overload can remove the highest observations and distort the upper tail.

Report station selection, measurement duration, season and missingness. The histogram alone cannot diagnose these design problems.

Relative frequency and frequency tell the same shape



Illustrative data or theoretical distribution. Exact plotted values remain editable.

How to read this graph

Dividing every count by the same n preserves the shape.

Relative frequencies facilitate comparisons with a sample of different size.

Different site coverage or seasons still prevent a fair comparison.

Unequal-width histogram calculation | Try it

A sample of 40 observations has 8 in [0,10), 12 in [10,30) and 20 in [30,80).

Calculate frequency-density heights and verify total area.

Unequal-width histogram calculation | Solution

Heights: 8/10=.8, 12/20=.6 and 20/50=.4 observations per unit.

Areas: .8×10=8, .6×20=12 and .4×50=20. Total area=40.

Probability-density heights divide these by 40: .02,.015,.01. Their areas total 1.

R laboratory: reproducible analysis

```
x <- c(12,18,21,24,27,29,32,35,38,42,44,47,51,54,58,63,68,72,85,96)
b <- seq(10,110,20)
f <- table(cut(x,b,right=FALSE))
cbind(f,relative=f/length(x),cumulative=cumsum(f))
hist(x,breaks=b,right=FALSE,xlab="PM2.5 (ug/m3)")
```

Expected frequencies: 6, 6, 5, 2, 1. Excel FREQUENCY uses a different boundary convention, so verify bin rules.

Excel laboratory: formulas and interpretation

Enter values in A2:A21. Lower boundaries go in C2:C6 and upper boundaries in D2:D6.

=COUNTIFS($A$2:$A$21,">="&C2,$A$2:$A$21,"<"&D2)

=E2/COUNT($A$2:$A$21) gives relative frequency. =SUM($E$2:E2) gives cumulative frequency.

Expected frequencies: 6, 6, 5, 2, 1. Excel FREQUENCY uses a different boundary convention, so verify bin rules.

Study references and further reading

OpenStax §§1.1–1.4, 2.1–2.2; NIST EDA §1.3.3; EPA QA/G-9S Chapter 2

OpenStax: free introductory explanations and exercises. NIST: formulas, assumptions and graphical diagnostics.

EPA QA/G-9S: environmental data quality and method-selection context. Penn State STAT 414: probability derivations beyond introductory coverage.

Course-list texts: Spiegel for worked calculations; Hogg & Craig or Mood, Graybill & Boes for mathematical development. Editions vary, so use the relevant topic index rather than an unverified page number.

Open-access reading links

OpenStax: https://openstax.org/books/introductory-statistics-2e/pages/1-introduction

NIST: https://www.itl.nist.gov/div898/handbook/

EPA: https://www.epa.gov/quality/guidance-data-quality-assessment

Penn State: https://online.stat.psu.edu/stat414/

R: https://stat.ethz.ch/R-manual/R-devel/library/stats/html/00Index.html

Excel: https://support.microsoft.com/en-us/excel/statistical-functions-reference